

## Home Assignment for Backend Developer (Java Spring)

### Objective:

Create a RESTful API for a simple task management system using **Java Spring**. The system should allow users to manage projects and tasks with user authentication via AWS Cognito. The code should be clean, well-documented, and follow best practices.

### Assignment Requirements:

#### Core Features:

1. **User Authentication:**
  - Implement user authentication using AWS Cognito User Pool (for this task, you can create these users manually).
2. **Project Management:**
  - CRUD operations for projects.
  - Each project should have a name and description.
3. **Task Management:**
  - CRUD operations for tasks within a project.
  - Each task should have a title, description, and status (e.g., todo, in-progress, done).

#### Additional Features:

- Include basic logging and error handling.
- Implement pagination in GET requests.
- Extra: Implement role-based access control (e.g., admin, user).

#### Testing:

- Write unit tests for core functionality using a testing framework

#### Submission Guidelines:

1. **Code Repository:**

Provide a link to a GitHub (or similar) repository containing the code.
2. **Deployment Suggestion:**

Provide a description of how you suggest this project should be deployed if it needs to handle 10k users a day and includes a client-side.

Good luck,